

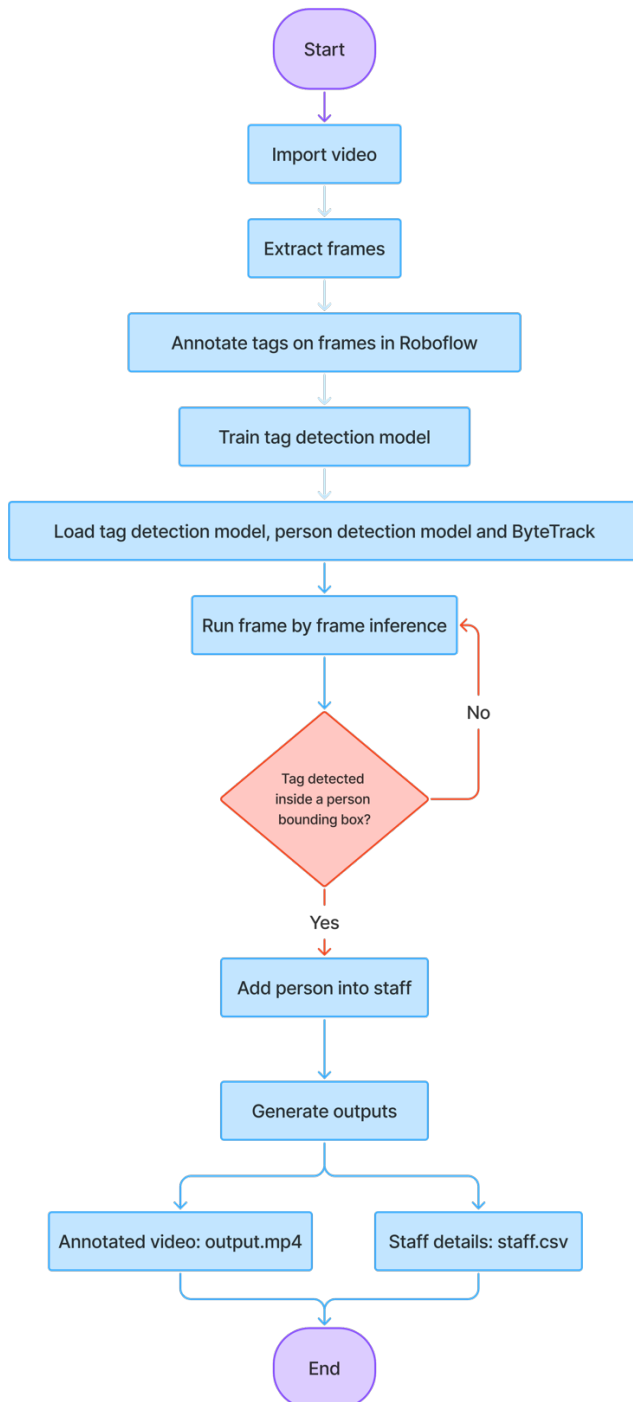
Staff Tag Detection

Detect and track person wearing a piano tag as staff in an office.

Objectives

1. Identify which frames in the clip have the staff present.
2. Locate the staff XY coordinates when present in the clip.

Flow Chart



Teck Stack

YOLOv8 model as a base model to fine tune person detection and piano tag detection. Then **ByteTrack** tracker is used to track person with piano tag. Besides, **Roboflow** is used to perform manual annotation on dataset. Lastly, **Python** is used as the only programming language.

Decision rule

A tracked person is labeled as staff only when all the following conditions are met:

1. Tag detection confidence is greater than 0.2
2. Tag is inside tracked person's bounding box
3. Tag is inside tracked person's bounding box for at least 5 frames

Challenges

YOLO models failed to accurately detect person subjects due to the use of fisheye lens in office's CCTV. Therefore, a new person detector model is trained using images of person under fisheye lens.

Staff often turn their backs to the camera or move behind objects, causing the tag or the person's bounding box to disappear. Which is an **occlusion** challenge. Thus, ByteTrack is implemented and set ByteTrack's track_buffer to 90 frames, therefore system can "remember" staff if staff is lost for several seconds.

Limitation & Future Improvement

The current model is heavily optimized for this specific video because the training images were extracted from the same source. This results in a high risk of overfitting. **To improve performance and ensure the model can generalize to other offices, future training should use a more diverse dataset featuring different subjects under fisheye distortion and piano tags from various locations.**

The current ByteTrack system is primarily limited to managing short-term occlusion. To enhance long-term tracking stability, **DeepSORT should be implemented to as it can memorize appearance-based features.** This approach enables better re-identification performance, especially when subjects remain hidden for prolonged periods.

Result

1. **Model Training result:** runs/detect/train/results.png and runs/detect/train-2/results.png
2. **Staff's details:** output/staff.csv
3. **Demo video:** output/output.mp4, <https://youtu.be/VM4Jzv85wOc>